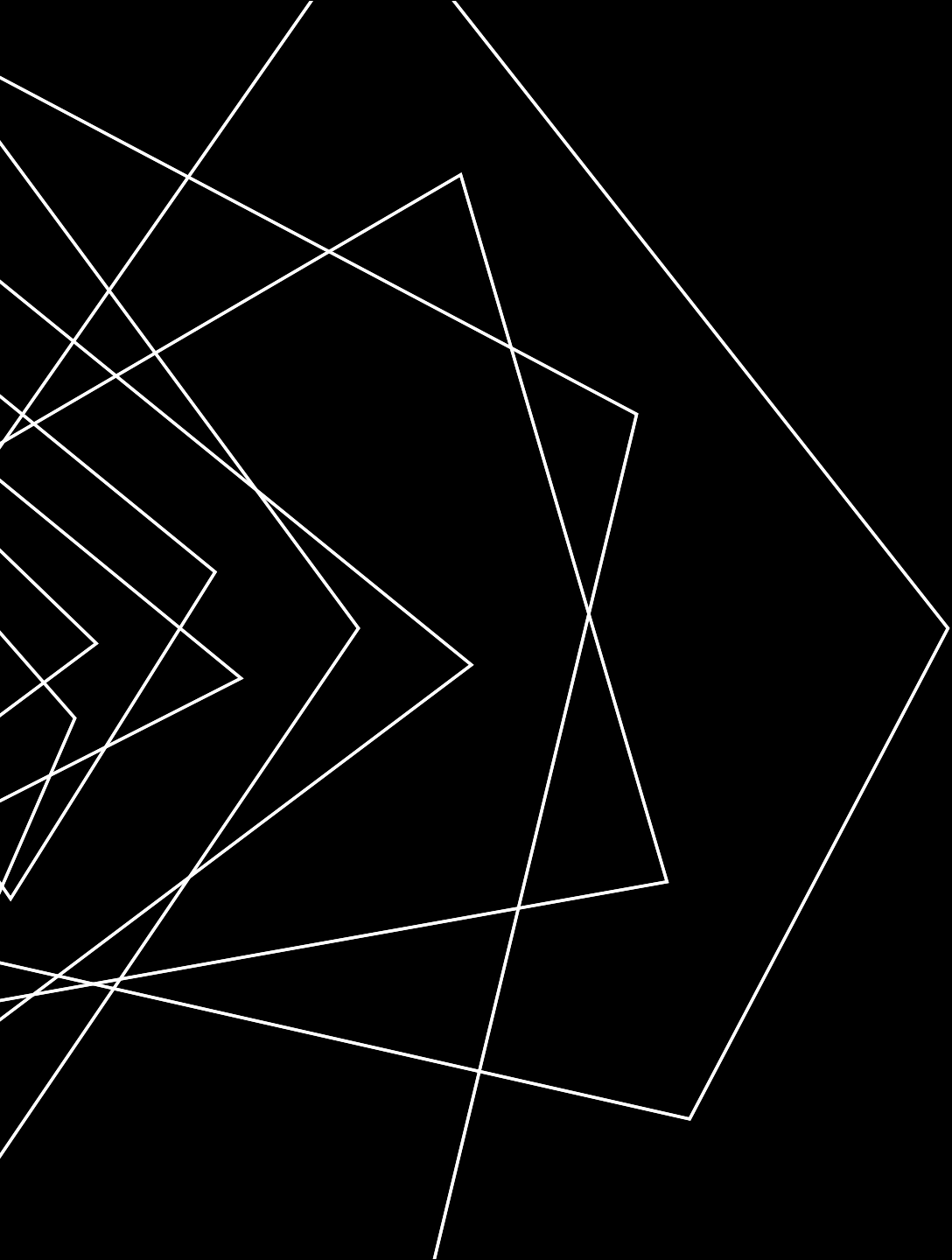




EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E
WEB PROGRAMMING
AND INTERFACE DESIGN



STATE, EVENTS, AND HOOKS

Web Programming and Interface Design

LEARNING OBJECTIVES

- Understand React state management with useState
- Handle user input and events
- Implement conditional rendering
- Build interactive UI components

WHAT IS STATE?

- State represents data that changes over time
- Managed within a component
- Example: Counter value, form input

USESTATE HOOK

- Import from React: `import { useState } from 'react';`
- Syntax: `const [state, setState] = useState(initialValue);`
- Example: `const [count, setCount] = useState(0);`

HANDLING EVENTS

- Use camelCase event names (onClick, onChange)
- Pass function as event handler
- Example: `<button onClick={handleClick}>Click</button>`

CONDITIONAL RENDERING

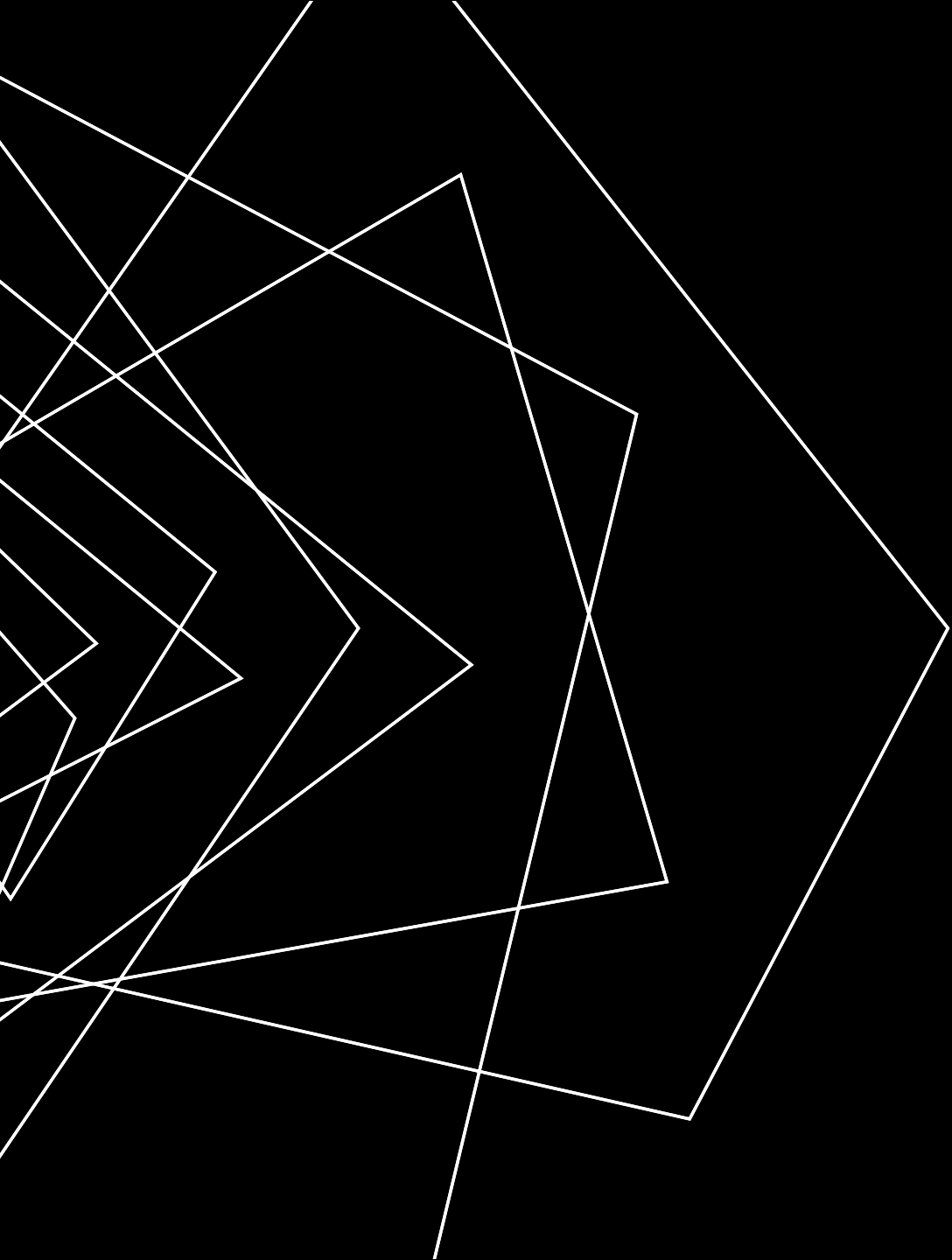
- Render elements based on state
- Use if, ternary operator, or &&
- Example: `{isVisible && <p>Visible!</p>}`

CODE EXAMPLE - COUNTER

```
function Counter() {  
  const [count, setCount] = useState(0);  
  return (  
    <div>  
      <p>Count: {count}</p>  
      <button onClick={() =>  
setCount(count+1)}>Increment</button>  
    </div>  
  );  
}
```

LAB EXERCISE

- Build a counter app with increment, decrement, and reset
- Create a toggle button to show/hide text
- Practice conditional rendering



LISTS, KEYS, AND USEEFFECT

Web Programming and Interface Design

LEARNING OBJECTIVES

- Render lists of data using `map()`
- Understand the role of keys in lists
- Use `useEffect` hook for side effects
- Fetch and display data from APIs

RENDERING LISTS

- Use JavaScript array `map()` method
- Example: `items.map(item => <li>{item}</li>)`
- Each element should have a unique key

KEYS IN REACT

- Help React identify which items change
- Should be unique and stable
- Example: `<li key={item.id}>{item.name}</li>`

USEEFFECT HOOK

- Runs side effects in components
- Syntax: `useEffect(() => { ... }, [dependencies]);`
- Examples: data fetching, subscriptions, timers

FETCHING DATA

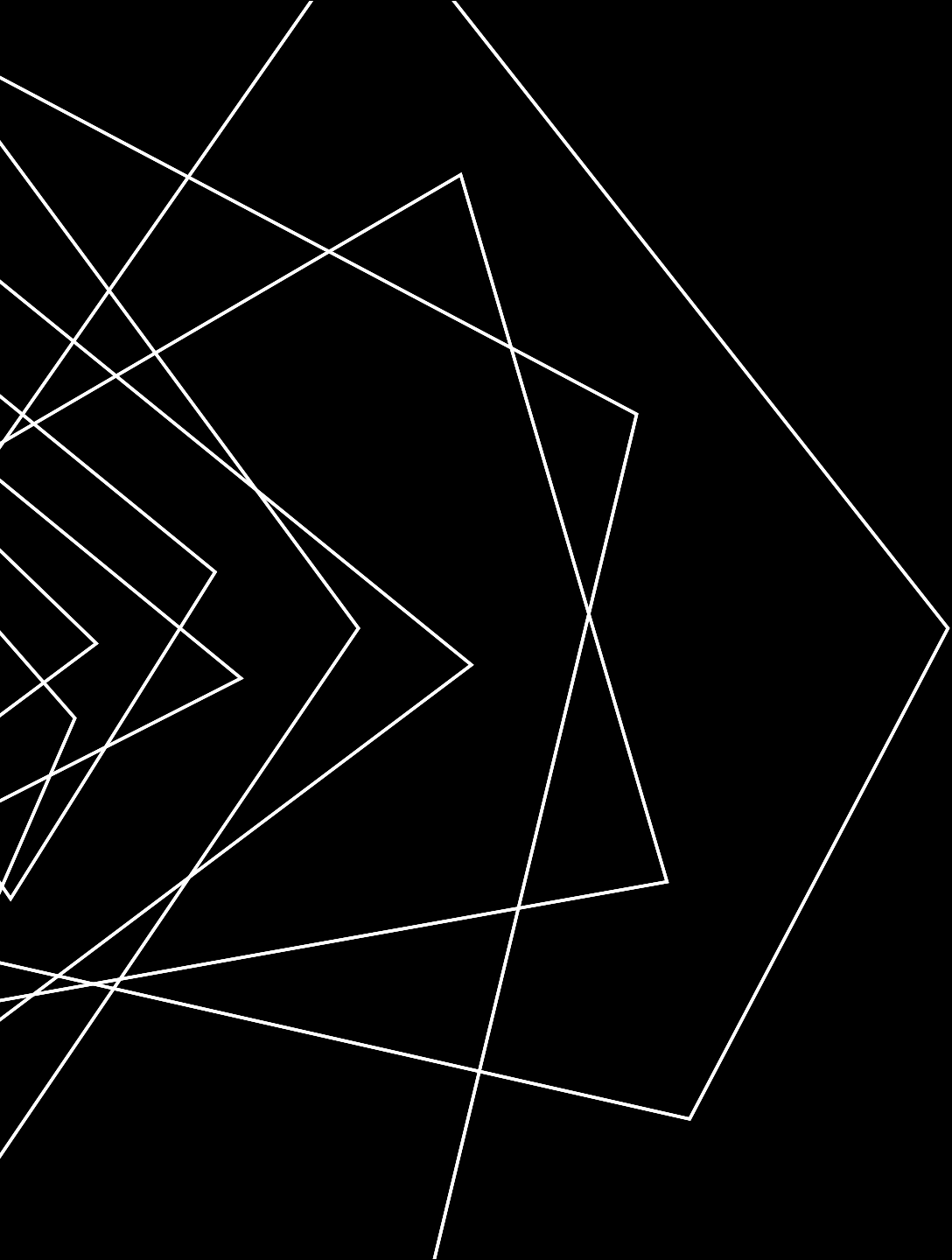
- Use `fetch()` or `axios` inside `useEffect`
- Handle loading and error states
- Example: `fetch('https://jsonplaceholder.typicode.com/posts')`

CODE EXAMPLE - FETCH POSTS

```
function Posts() {  
  const [posts, setPosts] = useState([]);  
  useEffect(() => {  
    fetch('https://jsonplaceholder.typicode.com/posts')  
      .then(res => res.json())  
      .then(data => setPosts(data));  
  }, []);  
  return posts.map(p => <p key={p.id}>{p.title}</p>);  
}
```

LAB EXERCISE

- Render a list of products using `map()`
- Add unique keys to each item
- Fetch posts from JSONPlaceholder API
- Display loading and error messages



THANK YOU

Yau Ka Cheung
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